

Universal Quantum Foam Hypothesis (UQSH)

An Ontological Framework for Field-Based Physical Interpretation

Ukshin Q. Rexhepi
Tübingen, Germany

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Abstract

This work is a conceptual and ontological framework. It does not propose new equations, numerical predictions, or experimentally testable parameters at this stage.

The Universal Quantum Foam Hypothesis (UQSH) posits a dynamic, continuous field medium in which physical quantities do not appear as sharply defined entities, but rather as stable state configurations. Uncertainty, dynamics, and stability are not disturbances, but fundamental conditions of field-mechanical organization. In this context, vagueness is understood not as an epistemic boundary, but as a functional property of stable states. States are not localized as points, but exist as spatially and temporally extended organizational forms of an underlying medium. Against this backdrop, gravity, matter, radiation, time, and temperature are interpreted as different manifestations of the same field. The UQSH interprets physical space as a continuous, dynamic field medium (Qu foam) that is constantly excited by radiation. Gravitation, electromagnetism, dark matter effects, dark energy, matter, and time are not conceived as independent fundamental entities, but rather as different states, properties, and dynamics of the same field. The approach is conceptual and speculative, but it does not contradict established experimental findings in relativity and quantum physics. The aim of this work is to formulate a coherent theoretical framework that brings together known physical phenomena from a field-mechanical perspective and reveals structural relationships between previously separate areas. The approach presented here differs explicitly from formalized unit field or origin field theories. The Qu-Foam hypothesis makes no claim to mathematical unification or the derivation of known equations, but rather understands itself as an interpretive framework that classifies existing empirical findings from a field-mechanical perspective without postulating new fundamental dynamics.

Central Introduction

Field, Uncertainty, and Organization

Modern physics describes natural phenomena with extraordinary mathematical precision. Nevertheless, the ontological meaning of many of its fundamental concepts, particles, waves, fields, space, time, mass, or temperature, remains open to interpretation. These concepts are formally clearly defined, but their physical meaning is interpreted differently depending on the theoretical context. The UQSH addresses precisely this openness to interpretation.

The UQSH understands the universe not as a collection of discrete objects in a passive space, but as a dynamic, continuous field medium. Stable structures do not arise spontaneously in this medium, but only under specific field-mechanical conditions. Matter, radiation, gravity, and quantum mechanical phenomena appear within this framework not as fundamentally separate categories, but as different forms of organization of the same field, realized in different saturation, dynamic, and coupling regimes.

A central tenet of UQSH is the departure from sharply defined entities in favor of viable states. Field states do not possess precisely localizable boundaries or completely fixed properties. They exhibit an intrinsic uncertainty, which is not understood as a measurement problem or epistemic limitation, but rather as a physical condition of their stability. In this sense, uncertainty is not a deficiency in description, but a constitutive feature of field-mechanical organization.

Dynamics, structure, and uncertainty are inextricably linked. Where uncertainty were completely suppressed, the field would lose its capacity for reorganization; where dynamics were completely relaxed, no stable patterns could be sustained. UQSH interprets stable forms of existence, from quantum mechanical states to material bonds and large-scale cosmic structures, as equilibrium states between field tension, dynamics, and unavoidable uncertainty.

The perspective proposed here does not aim to redefine established physical formalisms, but rather to provide a consistent framework for their significance. Quantum mechanics, relativity, and thermodynamic concepts retain their validity as precise descriptions of observable relationships. The UQSH offers a field-mechanical interpretive framework within which these descriptions can be understood as interconnected aspects of a dynamic field system.

In this sense, the UQSH is not conceived as a complete theory, but rather as a structuring interpretive framework. Its goal is not to provide definitive answers, but to illuminate the conditions under which concepts such as matter, time, temperature, or gravity can be meaningfully defined and related to one another in a physical sense.

Reader's Map (Reading Notes and Context)

How to Read This Text – Orientation for Readers from Established Physics

1. Aim and Nature of the Text

This text does not represent a replacement for the Standard Model of particle physics, general relativity, or established cosmology. It is intended as a conceptual framework for interpretation that reclassifies known, empirically confirmed phenomena within a unified field-mechanical language. All experimentally confirmed predictions of established theories remain unaffected. The aim is not to introduce new particles, new forces, or new dynamics, but to propose an alternative ontological reading of the same observable effects.

Readers should therefore not understand this text as a formal theory, but rather as a structuring perspective: It does not primarily ask how calculations are performed, but rather what physical quantities, fields, and particles might mean ontologically.

2. Ontological Shift: From Substances to Field States

The fundamental shift concerns the interpretation of physical entities. Concepts such as space, matter, radiation, gravity, or time are no longer treated as separate fundamental categories, but rather as different states of a single dynamic field medium.

This shift is deliberately formulated conservatively:

- Space remains geometrically describable,
- Radiation remains quantized,
- Fields retain their known equations,
- Conservation laws remain valid.

The only new aspect is the interpretation: Physical effects are not primarily attributed to additional substances (e.g., dark matter as particles), but rather to held or propagating field states.

3. Gravitation as a Continuous Field Reaction

A central guiding principle is that gravity does not "arise" only when matter is bound. Every field excitation, including free radiation, deforms the state of the spatial field, at least minimally. Gravitation thus appears as a continuous geometric reaction of the field, not as an effect coupled to a material threshold. In this context, matter differs from radiation not in principle, but functionally: Bound states fix field deformations locally and temporally, while free excitations carry these deformations along and allow the field to relax again.

4. The Electron as a Minimally Stable Field State

A central thesis, deliberately formulated cautiously, concerns the special role of the electron. It is not claimed to be the "only real particle," but rather interpreted as an empirically minimally stable bound field state.

This interpretation explains:

- its extraordinary stability,
- its universal role in atomic and chemical structures,
- its central importance for electromagnetic interactions.

Other particle types are not thereby negated, but understood as more complex or more tightly bound forms of organization within the same field. The thesis concerns stability and binding capacity, not the experimental existence of other particles.

5. Geometric Selectivity of Matter Formation

The text emphasizes that matter formation is extremely rare despite ubiquitous radiation. This is explained not primarily energetically, but geometrically. Stable binding requires specific constellations of field excitations in which several degrees of freedom, such as orientation, phase, functional role, and energy level, must be fulfilled simultaneously.

This geometric selectivity is key to understanding why very few patterns exist permanently and why matter is not continuously created anew.

6. Annihilation as a Field-Mechanical State Transition

Processes such as electron-positron annihilation are interpreted as the relaxation of bound field states. Crucially, this involves the complementary coupling of two mirror-image orientations of the same pattern. The resulting gamma radiation represents the return of previously held field work to free, propagating excitation.

This interpretation is fully compatible with experimental findings and avoids ontological assumptions about a physical "annihilation."

7. Cosmic Microwave Background Radiation as a Limiting Indicator

The cosmic microwave background radiation (CMB) is not interpreted as an energetic end state or as the driving force of cosmic dynamics, but rather as a long-lived limiting and circulating state of free field excitation in low-stress regimes. It is essential to understand that the CMB is not consumed. Along stress gradients, filamentary structures, or in strongly curved field regions, it can transition back into denser and higher-frequency regimes. Thus, the energy balance remains closed without requiring a single cosmological beginning.

8. Scales, Time, and the Speed of Light

The term "scale" is not used here to refer to additional spatial dimensions, but rather to describe different dynamic regimes of the same field. The speed of light appears as the maximum reaction and rotational speed of stable patterns within such a regime.

This interpretation neither relativizes local measurements nor violates the invariance of c . It serves solely to classify extreme time dilation, field saturation, and the apparent standstill of dynamic processes, such as those near black holes, within field mechanics.

9. Black Holes as Maximally Saturated Field States

Black holes are interpreted as maximally saturated states of the field, not as singular breaks. Their extreme stability arises from the fact that additional excitation can no longer be converted into internal dynamics. Structural analogies to stable ground states at other scales are explicitly meant conceptually and not literally.

10. Delimitation: How this text should not be read

- It does not claim to be a new experimental physics.
- It does not replace established equations.
- It does not introduce new particles.
- It is not a complete theory, but an open framework for working with the subject.

11. Invitation to Critical Examination

The UQSH is conceived as a testable, open framework. Its strength lies not in definitive answers, but in the systematic reorganization of known phenomena. Readers are expressly invited to critically examine, further develop, or specifically refute the proposed interpretations.

1. Introduction

Despite the great empirical successes of general relativity and quantum field theory, fundamental open questions remain concerning the nature of space, the cause of additional gravitational effects (dark matter effects), accelerated expansion (dark energy), the emergence of stable forms of matter, and the physical role of time.

These phenomena are predominantly treated separately in established physics, although they are structurally closely linked. The UQSH pursues a unifying approach and formulates a field-mechanical framework in which space, radiation, gravity, electromagnetism, matter, and time are understood as different states and manifestations of a single physical medium.

2. Core Hypothesis

Core hypothesis of the UQSH:

Physical space is not a passive background, but a continuous, dynamic field medium (Qu foam) whose state is influenced by radiation, energy, and matter. Gravitation, electromagnetism, dark matter effects, dark energy, matter, and time arise as distinct stable or metastable properties or states of this field, and not as independent fundamental substances.

This hypothesis:

- respects the local invariance of the speed of light,
- does not introduce any new fundamental particles,
- does not replace established field equations,
- interprets known physical phenomena in a field-mechanically unified manner.

3. Physical Interpretation of the Framework

3.1 Space as a Physical Field

Within the framework of the Universal Quantum Foam Hypothesis, physical space is not conceived as an abstract background or purely geometric structure, but rather as a physically effective, continuous field medium, referred to below as Qu foam. This Qu foam constitutes the fundamental supporting structure of all physical processes.

The Qu foam is locally deformable, elastic, and reactive. Field states can change spatially and temporally without requiring the assumption of discrete spatial cells or a strictly idealized continuum. A completely empty, unaffected state of space is not assumed, since the Qu foam is always a carrier of excitations, stresses, and relaxation processes.

In this understanding, space is not merely a stage for physical processes, but an active component of the dynamics themselves. Every interaction, every movement, and every radiation acts back on the state of the Qu foam and alters it locally or globally. The ontological status of space thus shifts from a passive geometric description to a physical medium with independent dynamic properties.

3.2 Radiation and Field Excitation

Within the framework of UQSH, radiation is understood as a physical excitation of the Qu foam that propagates through the field and interacts with its state. This view does not contradict established concepts: radiation remains quantized, carries energy and momentum, and follows known propagation laws.

UQSH complements this description with an explicitly field-mechanical perspective. Radiation is not considered as separate from its medium, but rather as a form of excitation that continuously influences the local state of the Qu foam. Every propagating radiation leaves a field-mechanical trace in the form of local and temporally limited deformation, stress, or excitation density.

Due to the ubiquitous presence of cosmic, electromagnetic, and thermal radiation, every region of space is in a permanently, albeit weakly, excited state. The Qu foam is therefore never perfectly still or homogeneous, but exhibits a dynamic interplay of excitation and relaxation at all scales.

This permanent field excitation forms a necessary field-mechanical prerequisite for later phenomena such as gravitational prestress, dark matter effects and the formation of bound material states.

3.3.1 Gravitation

Within the UQSH, gravity is interpreted as the geometric response of the Qu foam to any form of local field excitation and state change. Energy and matter alter the field state by generating local stresses and curvatures in the Qu foam.

Material bodies and radiation do not move due to a force acting at a distance, but rather follow the curved and strained structures of the field. Motion is thus a field-guided dynamic along energetically favored paths in the Qu foam.

The speed of light remains locally invariant relative to the respective observer. Scale-dependent effects do not concern the locally measurable speed of light itself, but rather the internal dynamics of field-mechanical states. Gravitation-induced effects such as deflection, retardation, or lensing effects do not arise from a change in the propagation speed, but from geometrically elongated, curved, or field-mechanically distorted propagation paths in the field.

This interpretation is fully compatible with the general theory of relativity, but supplements it with a field-mechanical reading in which spacetime curvature is understood as a physically supported state of stress of the Qu foam.

3.3.2 Inertia and Gravitation – A Field-Mechanical Interpretation of the Equivalence Principle

The equivalence principle states that inertial and gravitational mass are identical. While this identity is empirically established in classical physics and geometrically formulated in general relativity, its mechanical origin remains unexplained there.

Within the framework of the UQSH, inertia and gravity are understood as two manifestations of the same field-mechanical effect. Inertia arises from the necessary internal reorganization

of stable patterns during acceleration. Gravitation acts on the same patterns through spatially distributed field stresses that dictate preferred state trajectories.

In both cases, a coherent pattern must stabilize its internal structure under changing conditions. For the pattern, it is irrelevant whether this change is triggered by acceleration or by field geometry. From this follows the identity of inertial and gravitational mass as a necessary consequence of field mechanics.

Acceleration and gravity are therefore locally equivalent, since they trigger identical reorganization processes in the medium. The absence of negative gravitational mass results from the balancing nature of field tension. The equivalence principle thus points to space as a reactive medium in which structure, inertia, and gravity are jointly anchored.

3.4 Maximum Elasticity of Qu Foam

A key characteristic of Qu foam is its finite maximum elasticity. Field curvature and field stress cannot increase indefinitely. Once a critical load is reached, Qu foam no longer responds with further local linear deformation, but instead transitions into an altered field-mechanical state in which additional stress does not lead to further compaction, but rather to a spatial expansion of the boundary region.

This elastic limit represents a fundamental field-mechanical constraint and prevents diverging field quantities. It forms the basis for:

- the formation of stable, bound field states (matter),
- the limitation of extreme gravitational compaction,
- the existence of compact objects, including black holes.

In this context, black holes are not interpreted as singular points, but rather as local limiting states of maximum field saturation of the Qu foam, where maximum elasticity is reached and a field-mechanical regime change occurs. This regime change affects only the directly impacted field region and represents neither a break in field continuity nor a global state of the Qu foam.

As the size of a black hole increases, the region in which the Qu foam operates at its elastic limit expands. This increases the elastic restraint of the surrounding field over larger scales. The growth of compact objects thus remains possible in principle, but is not abruptly limited; instead, it is progressively hindered by the increasing restraint of the surrounding field.

The maximum elasticity of the Qu foam therefore acts as a stabilizing, regulating mechanism that allows local extreme states without permitting an uncontrolled collapse of the field. It enables the formation of long-lived, structurally stable configurations at all scales, from elementary bound states to cosmic limiting objects.

3.5 Time as a Field-Mechanical Property

In UQSH, time is not introduced as an independent fundamental dimension, but rather as a field-mechanical emergent phenomenon that arises from the dynamics of the Qu foam. Time is not understood here as an independent background parameter, but as a measure of the changeability of states. Different degrees of organization lead to different relaxation and development times. Time describes the ordered sequence, duration, and causal structure of field state changes.

A completely static, unchanging field state would not possess a physically definable time. Time arises where field states change, relax, bind, or reorganize. It is thus inextricably linked to dynamics and interaction.

Measurable proper time along a world line remains a well-defined and experimentally confirmed physical quantity and is interpreted as a local measure of field-mechanical state evolution. In this view, gravitational time dilation results from an increasingly bound or slowed field-mechanical relaxation in highly strained regions of the Qu foam.

The arrow of time arises from the asymmetric irreversibility of certain field processes, particularly dissipation, binding, and stabilization of field structures. These processes generate a preferred direction of field development without requiring the postulation of a fundamental direction of time.

3.6 Electromagnetism as Structured Field Excitation

Within the framework of UQSH, electromagnetism is understood as a highly structured, coherent form of field excitation of the Qu foam. Electromagnetic fields correspond to coherent states of the field with characteristic polarization, phase, and propagation direction.

Electric charge is interpreted as a local, directed asymmetry of bound field nodes, which determines how these nodes couple with electromagnetic excitations. Electromagnetic interactions arise from the directed field-mechanical coupling between free and bound excitation states of the Qu foam.

Unlike gravity, which occurs as a scalar or geometric reaction to any field excitation, electromagnetism is selective and direction-dependent. It only acts where bound field states possess a corresponding orientation or coupling capability.

This perspective allows for a unified classification of electromagnetic phenomena within the field-mechanical framework without replacing or modifying the established electromagnetic equations. Rather, they are understood as effective, highly precise descriptions of specific organizational forms within the field.

Interim conclusion

With the field-mechanical interpretation of space, radiation, gravity, elasticity, time and electromagnetism, a consistent physical interpretive framework has been established, from which the further consequences of UQSH, in particular dark effects and the emergence of stable states of matter, follow directly.

4. Consequences of the Framework

4.1 The Dark Matter Effect as a Key Component of the Core Hypothesis

The dark matter effect is not a secondary addition within the UQSH, but rather a direct consequence of the core hypothesis. It follows from the assumption that the Qu foam, as an elastic field medium, can develop large-scale stress and curvature states that do not fully relax.

These field-mechanical prestresses arise from the long-term interaction of:

- permanent field excitation by radiation,
- the gravitational effect of baryonic matter,
- the nonlinear superposition of many weak field deformations.

The Qu foam can stably support such prestressed states without the need for bound matter. These states are gravitationally active and generate additional field curvature, which manifests itself observably in galactic rotation curves, gravitational lensing, and large-scale structure formation.

In this context, the dark matter effect is not interpreted as evidence of missing particles, but rather as an intrinsic property of the space field itself. It is therefore not an additional postulate, but a fundamental component of the UQSH, which follows directly from the field-mechanical dynamics of an elastic Qu foam.

Dark matter-like effects manifest in various observable phenomena, including the flattening of galactic rotation curves, gravitational lensing effects, and systems such as the Bullet Cluster. The radial acceleration relation shows a close coupling between the acceleration calculated from baryonic matter and the actually observed dynamics, while lensing measurements often yield a stronger deflection of light than can be deduced from the visible mass alone. In the Bullet Cluster, there is also a spatial separation between hot baryonic gas and the centers of lensing.

Within the framework of the Universal Quantum Foam Hypothesis, these findings are not interpreted as evidence for an additional, independent matter component, but rather as field-mechanical consequences of an elastic space field. The additional gravitational effect arises from nonlinear stress states of the Qu-foam, which are generated by bound baryonic structures. While stable, coherent structures such as stars and galaxies largely retain their field stress state even in dynamic processes, highly turbulent states such as hot gas lose much of their coherent field effect. Light propagates along the geodesics of this physically stressed medium and reacts to the integrated field geometry, resulting in enhanced deflection and apparent mass decoupling without the assumption of a separate dark matter component.

A more detailed, problem-oriented discussion of individual dark matter anomalies (RAR, bullet cluster, core-cusp, etc.) can be found in an accompanying monograph by the author, published via Amazon KDP.

4.2 Dark Energy, Field Relaxation, Voids, and Higher-Order Redshift

Within the framework of UQSH, dark energy is understood solely as the field-mechanical relaxation tendency of the Qu foam. In low-stress, structurally weak regions, the fundamental field stress can unfold more freely. Dark energy is thus not a substance or an independent agent, but rather a description of a scale and regime state of the field medium.

The observed redshift is strictly distinguished from this. It arises from the propagation of radiation patterns in the Qu foam. Over large distances, the effective wavelength changes because the radiation excitation is distributed over increasingly larger spatial fractions, and its coherent field structure is smoothed field-mechanically during propagation.

However, this fundamental effect is not homogeneous. In voids, the coupling and disturbance density of the field is lower: radiation patterns are rearranged less, disturbed less, and therefore appear to have longer wavelengths. This results in a section-by-section increase in the observed redshift in void passages, without the dark energy itself causing the redshift.

Cosmological lines of sight thus exhibit a sequential superposition of field-mechanical regimes: normal field propagation in structured regions → void passage (enhanced effect) → normal propagation again → next void passage, and so on. The radiation arriving at the observer therefore carries a cumulative, path-dependent redshift of a higher order, resulting from the repeated traverse of differently strained field regions.

Current tensions between local and large-scale cosmological distance and expansion measurements can be interpreted in this context as an indication that cosmological lines of sight are influenced to varying degrees by voids and structure-rich regions, and that path-dependent contributions to redshift can be physically relevant. Dark energy appears here as a local field-mechanical relaxation state that influences the large-scale structure of the field without itself causing the redshift or manifesting as an independent dynamic quantity.

4.3 Blueshift and Redshift as Field-Mechanical Density Phenomena

Within the framework of UQSH, the propagation of radiation is not only understood as the movement of energy, but also as the field-mechanical distribution of excitation probability in the Qu foam. Even at the point of emission, radiation is not localized as a point, but possesses a spatially distributed excitation structure whose local density changes continuously along its propagation path.

With increasing distance from the emission point, the same radiative excitation is distributed over an increasing volume of the Qu foam. This results in a decrease in the excitation density per volume element. This fundamental rarefaction is a general characteristic of radiation propagation and occurs independently of large-scale cosmological effects.

In this context, the observed frequency of radiation is directly coupled to the local field-mechanical density of the excitation. Changes in the effective wavelength result from the degree to which the excitation probability is spatially concentrated or distributed, not from a separate, independent process.

Blueshift occurs where radiation excitations become more densely packed due to field mechanics. This happens in regions with strong field curvature or compression, such as near massive or compact objects. The increased excitation density leads to shorter effective wavelengths and higher observed frequencies. Blueshift is thus an expression of a local increase in radiation density in the Qu foam.

Redshift arises where radiation patterns in the Qu foam become more dilute than would be expected from purely geometric propagation. This can occur through large-scale field expansion, field-mechanical relaxation, or—as explained in Section 4.2—through repeated traverses of low-stress regions. The reduced excitation density causes a lengthening of the effective wavelengths and lower observed frequencies.

In this interpretation, blueshift and redshift appear as two sides of the same field-mechanical principle. They reflect local or large-scale changes in radiation density in the Qu foam and do not represent independent, separate processes.

This perspective complements the kinematic description of frequency shifts with a physical media theory in which frequency changes are understood as a direct consequence of the field-mechanical distribution and restructuring of radiation excitations. In this context, dark energy is understood as a local effect.

4.4 Temperature as a Field-Mechanical State Variable (UQSH)

Within the framework of the UQSH, temperature is not understood as an independent fundamental quantity, but rather as a macroscopic indicator of the dynamic state of a field regime. Temperature does not describe energy per se, but rather the effectiveness of field-mechanical dynamics per available degree of freedom.

A field state possesses energy regardless of whether this energy is dynamically effective or exists as a sustained field stress. Temperature therefore does not measure the absolute energy content, but rather the portion of the energy that manifests itself in cycles, oscillations, and reorganization processes. The more dynamics a field state can sustain, the higher its temperature appears.

In this understanding, temperature is not a primary parameter, but a derived state parameter that indicates how freely excitations within a field can reorganize. It is thus closely coupled to the local field elasticity, the geometric constraint, and the degree of stress binding.

4.5 Temperature and Field Tension

Field tension and temperature are not identical, but are indirectly related. Field tension describes the degree of local curvature, prestress, and saturation of the Qu foam. Temperature indicates how much of this tension is still converted into dynamic motion.

- At low field tension, excitations can circulate freely → low, but defined temperature.
- At medium field tension, circulation is forced and stabilized → high temperature.
- At extreme field tension, circulation is increasingly suppressed → temperature loses its physical significance.

Temperature thus does not measure the magnitude of the tension itself, but rather the mobility under tension.

4.6 Cosmic Temperature and the “Cold” Universe

The large-scale universe is predominantly in a state of extremely low local field density. Matter density, field curvature, and reorganization rates are minimal. The Qu foam operates here near its minimally excited state.

In the UQSH, the cosmic microwave background radiation, with an effective temperature of about 2.7 K, does not mark a thermodynamic end state or a mere remnant, but rather the lowest dynamically supported field state in which a thermal description is meaningful at all. Below this level, field excitations exist, but no statistically relevant cycles exist that would allow a temperature definition.

The universe is therefore not “cold” because of a lack of energy, but because of a lack of dynamics.

4.7 Local Warming and Planetary Temperatures

Regions like Earth exhibit significantly higher temperatures because they present highly dynamic field states locally. Several factors interact:

- local field compression due to mass,
- continuous input of radiation (e.g., from the sun),
- geometric constraints imposed by material structures, the atmosphere, and surfaces.

These conditions prevent complete field relaxation. Energy remains bound in dynamically active degrees of freedom, thereby generating and maintaining temperature. Here, temperature is an expression of sustained dynamics, not merely energy concentration.

4.8 Temperature in Voids

In the UQSH framework, cosmic voids represent regions of maximum field relaxation. Matter density, field curvature, and dynamic reorganization processes are minimal there. Accordingly, voids should exhibit the lowest physically measurable temperatures.

At the same time, below a certain dynamic level, temperature loses its operational significance. Voids are therefore not simply "colder," but rather approach a state in which thermodynamic concepts become increasingly meaningless.

4.9 Temperature in Extremely Saturated States

Near black holes, the observable temperature initially rises sharply because field excitations are accelerated, focused, and reorganized. However, this effect only holds true as long as dynamics are still carried.

With increasing field saturation, orbits gradually transition into carried field tension. At the limit of maximum saturation, temperature completely loses its physical meaning. There is neither a state of infinite nor zero temperature, but rather a regime in which temperature as a state variable is no longer defined because no dynamic degrees of freedom exist to which it could refer.

4.10 Classification

In the UQSH, temperature does not appear as a fundamental property of matter, but rather as a field-mechanical state variable that indicates how much dynamics a field regime can currently support. It is thus closely linked to field tension, geometry, and reorganization capacity.

Temperature does not measure how much energy is present, but rather how much movement is possible.

4.11 Compression, Desaturation, and Re-emission of Radiation in Extremely Saturated Field States

In highly saturated field states such as neutron stars and black holes, incoming radiation undergoes pronounced field-mechanical compression. Free-radiation excitations entering these regions pass through areas of greatly increased field stress and curvature, causing their excitation density to increase continuously. This process should not be understood as an abrupt event, but rather as a gradual geometric focusing within an increasingly saturated field regime. The effective frequency of the radiation increases with the local field compression, as the same excitation energy is concentrated on an ever smaller field-mechanical volume.

Within the framework of UQSH, this densification cannot proceed indefinitely. If the field locally reaches its maximum elasticity or saturation limit, additional incoming excitation is no longer integrated into internal dynamics. Instead, the field state transitions into a reorganization regime in which the previously focused excitation is converted into higher-frequency, freely propagating radiation modes. The saturated field state thus does not act as an energy sink, but rather as a transformation and deflection zone in which radiation transforms its incoming density and frequency depending on the field. The exit from such regions in the form of high-frequency emission—for example, in the gamma or higher spectrum—does not represent an energy generation process, but rather a field-mechanical return of previously bound or focused excitation to a freely propagating state.

In this view, neutron stars and black holes function as extreme compression and reorganization zones of the Qu foam. Despite high field stress, neutron stars still possess limited internal dynamics and can both focus and re-emit radiation. Black holes represent the limiting case of maximum field saturation, in which incoming radiation is largely integrated into the carried field stress and only reverts to free radiation via indirect, field-mechanically mediated transitions. Processes such as Hawking radiation appear in this context as an extremely weak, continuous limiting case of such field-mechanical re-emission at transition surfaces between saturated and less saturated regimes.

Under progressive field compression, even bound material field patterns lose their stability. This affects not only complex structures but also elementary bound states such as electrons. When an electron pattern is subjected to increasing field saturation, the excitation density of its internal organization rises continuously. If the local field voltage exceeds the stability limit of this pattern, the bound organization can no longer be maintained. The electron is not lost in this process but rather dissolves into higher-frequency, free radiation modes. Matter thus appears not as an unchanging substance but as a metastable state of field organization that can reversibly transition into radiation under extreme conditions.

Conversely, this connection also provides a natural avenue to understanding matter formation. In the transition regions of highly saturated field states—for example, in the vicinity of neutron stars or in the outer regimes of black holes—high-frequency radiation modes, particularly in the gamma spectrum, can re-enter regions where geometric focusing, phase coupling, and field-mechanical selectivity allow for stable binding. Within this framework, matter formation does not appear as a primary process, but rather as a highly selective feedback process within a circulating field system, in which radiation, bound patterns, and saturated states continuously transition into one another.

Gamma radiation thus assumes a dual role: It represents both the end product of field desaturation under extreme compression and a potential starting point for new material binding states. The energy balance remains closed. Energy is neither created nor destroyed, but rather alternates between free radiative excitation, focused field work, and bound pattern

organization. In this view, extreme astrophysical objects do not appear as special cases, but as active nodes of a field-mechanically circulating system.

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4.13 Incoming Radiation and Luminosity of Massive Compact Objects

The extraordinary luminosity of quasars and active galactic nuclei is predominantly interpreted in standard astrophysics as a consequence of accretion: matter falls into a deep gravitational potential, is heated, and emits radiation. This description captures the energetic aspect of the process but leaves open the question of why the regions immediately in front of the event horizon develop extremely luminous, highly variable, and long-lived radiation sources.

Within the framework of the UQSH, this finding is interpreted in terms of state dynamics. The region in front of a massive compact object represents an extreme regime of the medium in which gravitational field tension, high particle densities, strong magnetic fields, and relativistic velocities interact. Incoming radiation does not encounter an empty background there, but rather an already highly excited and highly coupled state field.

Due to gravitational deflection and time dilation, photon trajectories near the event horizon are lengthened and spatially concentrated. For an external observer, this leads to an effective accumulation of incoming radiation in a thin pre-horizon zone. In this zone, the local radiation density increases sharply, causing interactions between radiation modes to become increasingly nonlinear. Here, radiation functions not only as an energy carrier but also as a carrier of state information that can couple phase-wise.

When the local density of states reaches a saturation threshold, classical outflow channels—continuous radiation and thermal emission—can no longer be maintained linearly. In UQSH, saturation is understood as the trigger for state transformations: The medium responds to excess coupling by forming new coherent carrier states. At the known physical level, this manifests as pair production, dense radiation plasmas, and strongly coupled particle-radiation regimes. These processes lead to a portion of the incoming radiation being reorganized into material states.

The newly formed states are gravitationally bound and contribute again to the density of states in the forehorizon region. This creates a feedback loop: Incoming radiation increases local excitation, saturation forces state transitions, and the resulting matter in turn feeds the highly excited region. In this view, the observed extreme luminosity of quasars is not explained solely by a single accretion, but by a self-organizing cycle of incoming radiation, state transitions, and renewed emission of radiation.

Supermassive black holes are particularly well-suited for the formation of such regimes because their forehorizon regions are spatially extended and allow for long effective residence times for incoming radiation. The UQSH therefore interprets quasars not as mere end products of matter inflow, but as stable, highly excited state regions in front of the event

horizon, where radiation, matter, and field tension are continuously transformed into one another.

This interpretation is consistent with observations of high luminosity, rapid variability, and the close coupling of radiation and jets in active galactic nuclei, without requiring a violation of established relativistic or quantum physical principles.

4.14 Application to the Electromagnetic Spectrum

Within the framework of UQSH, the electromagnetic spectrum is understood not as a collection of ontologically distinct types of radiation, but as a continuous scale of field-mechanical excitation states of the same physical radiation pattern in the Qu foam. Differences between radio waves, microwaves, visible light, X-rays, and gamma rays arise from the local excitation density, coupling strength, and field curvature, not from qualitatively different entities.

Long-wavelength radiation corresponds to states of low field-mechanical excitation density, in which radiation patterns are distributed over large volumes and couple only weakly with the Qu foam. Short-wavelength radiation occurs where radiation excitations are highly concentrated, dense, or geometrically focused, for example, near compact objects or in regions of high field potential.

In this understanding, the electromagnetic spectrum forms a density- and state-dependent manifestation scale of the same fundamental physical structure. Transitions between different frequency ranges are continuous and reflect changes in the field-mechanical state, not the switching between different types of radiation.

This interpretation unifies blueshift and redshift, high-energy emissions, and low-frequency radiation within a common field-mechanical framework and situates the electromagnetic spectrum as an expression of the local and large-scale dynamics of the Qu foam.

Within the framework of UQSH, the observed colors and intensities of radiation are not exclusively field-dependent but result from the interplay between the source and field-mechanical propagation. The Qu foam simultaneously carries a multitude of different radiation patterns originating from various sources and differing in frequency, intensity, and internal structure.

The field-mechanical properties of the Qu foam do not generate radiation but rather modulate it: they influence how existing radiation patterns propagate, condense, or rarefact. Blueshift and redshift thus alter the effective wavelength, excitation density, and coherence of existing radiation without eliminating its source-specific characteristics.

The continuous color scale of the electromagnetic spectrum should therefore not be understood as a direct representation of a single field state, but rather as a visible expression of the superposition of many source-specific radiation patterns, which are modulated differently along their propagation path.

4.15 Matter as a Bound Field State (Geometrically Selective)

Within the framework of the UQSH, matter is understood as a locally bound, stable state of the Qu foam. It is not a substance added to space, but rather a specific organizational form of the field itself. The transition from radiation to matter occurs where free field excitations, due to the finite elasticity of the Qu foam, can no longer fully relax and instead transition into a permanently supported state.

Free radiation excitations propagate unbound through the field as long as local stresses can be reduced by elastic relaxation. If this capacity is exceeded—for example, through superposition, geometric focusing, or long-term prestressing—a locally bound field structure stabilizes, remaining anchored in the Qu foam. Matter thus does not represent a categorical break, but rather a field-mechanical regime change from free to bound dynamics.

This stabilization is not solely a matter of energy or intensity, but is subject to strict geometric selectivity. The UQSH assumes that bound field states can only arise if several free excitations form a configuration that allows for a closed, self-supporting field-mechanical organization. The minimum necessary condition for this is triadic coupling: Three radiation excitations must superimpose in such a way that they are neither linearly propagated nor mutually annihilated, but rather form a rotatable, closed field node.

Two superimposed excitations are insufficient in this model, as they tend either toward propagation or destructive interference. Four or more excitations, on the other hand, lead to overdetermined or unstable configurations that do not permit permanent binding. Triadic geometry thus represents the minimum necessary geometric condition for the emergence of stable elementary states of matter.

The commonly used description of a 3×3 structure does not represent an extension beyond this triad, but rather describes the stabilized final state of a single triadic field node. It refers to the complete development of three intertwined freedom domains within the same triad:

- (1) functional roles of the bound pattern,
- (2) stable geometric orientations,
- (3) energetic levels of organization.

Only when these three aspects are realized simultaneously does a permanently self-sustaining vortex state arise in the Qu foam. Incompletely developed constellations relax back into free radiation or remain as metastable transitional forms. Within this framework, the extraordinary stability of elementary matter forms does not result from their energy, but from the rare geometric and field-mechanical completeness of their internal organization.

Matter formation is therefore possible in principle, but highly selective. Only a very small fraction of all radiation superpositions simultaneously fulfill the necessary geometric, energetic, and field-mechanical conditions. This explains both the low matter density relative to the ubiquitous radiation and the dominance of free radiation as an information carrier over large distances, without contradicting the conservation laws of standard physics.

Crucially, the radiation relevant for matter formation is not bound to a specific cosmological origin. Free-field excitations from any source—local, galactic, or large-scale—can, in principle, enter the transition from free to bound dynamics, provided the triadic coupling and saturation conditions are met. Long-wavelength, highly coherent radiation does not represent a privileged matter driver, but merely one possible initial state within a broad spectrum of free-field excitations.

4.16 The Electron as the Elementary Pattern Knot of Matter

Within the framework of the Universal Quantum Foam Hypothesis, matter is not understood as an accumulation of ontologically distinct particles, but rather as the result of stable pattern bonds within a dynamic field medium. In this view, the electron occupies a special position: it does not appear as one of many equally ranked fundamental particles, but as the simplest permanently stable bound pattern knot upon which all higher-order matter structures are built.

The empirically striking universality of the electron—its presence in all atoms, its role in chemical bonds, its participation in electromagnetic interactions, and its stability over cosmic timescales—suggests that it is not a random entity, but rather a field-mechanically favored binding state. Other known forms of matter appear in this perspective not as independent basic building blocks, but as more complex interconnections, entanglements, or saturation states of this fundamental electronic pattern.

The formation of such a stable electron pattern is explained in UQSH not primarily energetically, but geometrically. Free radiation excitations can propagate unbound within the Qu foam as long as local stresses can be relieved by elastic relaxation. Only under specific constellations—especially a triadic superposition of radiation patterns—does a regime change occur. In an approximately trigonally symmetric geometry (relatively around 0° , 120° , and 240°), the excitation is neither linearly propagated nor annihilated, but rather transformed into a closed, rotatable field node.

This closed rotation is not arbitrary, but structurally linked to π : The stable electron state corresponds to a minimal, fully rotating field vortex whose dynamics involve a complete phase rotation. π appears here not as a numerical constant in the strict sense, but as an expression of the necessary completeness of a closed rotation, without which no permanently bound state can arise.

Triadic coupling represents the minimum necessary condition for binding. Two superimposed excitations typically lead to transmission or mutual cancellation, while four or more couplings tend toward overdetermined, unstable field configurations. Only three-way coupling allows for a rotatable, self-supporting structure.

However, the long-term stability of the electron does not arise solely from this three-way coupling, but from a more extensive 3×3 selectivity. This does not describe additional dimensions, but rather the minimum complete organization of the bond: three functional roles (e.g., setting, modulation, damping), three geometric orientations, and three energy levels. Only when these degrees of freedom are all fulfilled does an electron emerge as a long-lived pattern node.

In this context, ϕ (the golden ratio) plays a mediating role. It does not appear as an exact specification, but rather as a stabilizing class of proportions that allows the system to remain permanently bound despite unavoidable uncertainty. In this sense, the electron's field geometry is neither exactly symmetrical nor arbitrarily variable, but rather operates within a preferred stability window that permits self-similarity across scales and minimizes disturbances. Here, ϕ describes the optimal balance between order and flexibility necessary for a pattern to remain both stable and capable of connection.

From this perspective, the uncertainty observed in quantum physics is not an obstacle to matter formation, but rather its prerequisite. Exactly defined, rigid configurations would be mechanically unstable. Only a limited geometric range allows the electron pattern to absorb disturbances without losing its binding. Uncertainty thus appears as a functional property of stable field organization.

Annihilation—classically described as the "annihilation of matter"—is interpreted within the UQSH framework as field relaxation during complementary coupling. When an electron and a positron collide, two mirror-image orientations of the same pattern node couple. Their field potentials allow for complete relaxation of the bound state. The gamma radiation observed in this process does not represent "residues" of particles, but rather the return of previously held field work to free, propagating excitation. The number of emitted gamma quanta results from the internal geometry of the coupled state and reflects different relaxation pathways.

From this perspective, protons and nuclei do not appear as ontologically more fundamental objects, but rather as high-energy, locally tightly bound patterns whose stability arises from the coupling of many fundamental electronic forms. Antimatter experiments do not contradict this view: The field allows complementary orientations of the same fundamental pattern, but generates them predominantly transiently. In an environment dominated by electronic matter, positronic states preferentially relax back into free radiation.

Overall, this approach suggests that, in a field-mechanical sense, there exists a minimally stable fundamental form of matter: the electron as the self-supporting pattern node of the Qu foam. All other known forms of matter—from atomic nuclei to molecules and macroscopic structures—can be understood as hierarchical binding, entanglement, and saturation states of this fundamental pattern. Properties such as spin, charge, mass, and the known interactions appear as binding modes and geometric degrees of freedom within a unified field-mechanical system.

This shifts the fundamental ontological question of physics: not "how many types of particles exist," but rather which pattern geometries are field-mechanically stable. That the electron uniquely fulfills this role in the UQSH is not a coincidence, but the direct consequence of triadic coupling, limited field potential, closed rotational dynamics (π), and stabilizing proportional windows (ϕ).

4.17 The Cosmic Microwave Background (CMB) as an Observable Limiting Indicator

Within the framework of the UQSH, the cosmic microwave background (CMB) is not interpreted as a remnant of a singular cosmological initial state, but rather as a long-lived limiting and circulating state of free-field excitation within the quantum foam. It marks a regime of extremely low excitation density in which the field operates large-scale near its minimum stress states without fully relaxing or transitioning into a static state.

In this understanding, the CMB represents an extremely long-wavelength, highly coherent mode of free-field excitation. Its low local excitation density indicates that this radiation preferentially propagates stably in low-stress field regions. At the same time, it remains field-guided and dynamic, thus clearly distinguishing it from a fully relaxed or energetically "spent" state. The CMB is therefore not a fading residue and not a thermodynamic endpoint, but a stable limiting indicator of the current field-mechanical state.

Crucially, this radiation is not considered energetically lost within the UQSH framework. Energy does not change its magnitude, but rather its state. Along stress gradients, large-scale field structures, or near highly curved and compressed field regions, longwave background radiation can be transformed back into denser and higher-frequency regimes. This transition occurs through geometric focusing, field-mechanical compression, and nonlinear coupling within the Qu foam. It is not an energy generation process, but rather a redistribution of energy between free propagation, focused field work, and—under highly selective conditions—also into bound material states.

The cosmic microwave background radiation thus does not mark a thermodynamic standstill, but rather a circulating state within a circulating, energy-conserving field system. Radiation can transition between minimally excited, low-stress regimes and higher-energy states without requiring a global energy loss or a single cosmological initial state. Material bonding is not necessary for this rearrangement and represents a separate, geometrically highly selective process.

Several observable consequences arise from this interpretation. The physical significance of the CMB lies not primarily in its mean temperature, but in deviations from ideal isotropy, perfect coherence, and structural uniformity. Correlations between CMB anisotropies and large-scale field and structural properties, such as filaments, voids, or galaxy clusters, can be consistently interpreted as expressions of ongoing field-mechanical interactions. Directional or structure-dependent variations in coherence provide evidence of dynamic transitions between low-stress and focused field regimes in the present universe.

From this perspective, the CMB functions as a continuously embedded measurement of the current field state and not as a relic of a closed cosmological epoch. It allows inferences about large-scale field stress, relaxation dynamics, and rearrangement processes within the Qu foam without requiring the introduction of additional substances or singular initial assumptions.

The role of the CMB as an observable limiting indicator complements the field-mechanical interpretation of dark matter, dark energy, and matter formation within the UQSH. It closes the global energy balance of the model by understanding radiation as a permanently carried component of a dynamic field system in which energy can circulate between minimal excitation, focused compression, and higher-frequency emission.

5. Scale-Dependent Dynamics of the Space Field and the Special Role of Stable Pattern States

5.1 Continuous Field Reaction Instead of Threshold-Like Gravitation

In standard physics, gravity is often described as the effect of massive objects, while radiation is primarily considered a carrier of energy and momentum, whose gravitational influence appears negligibly small in everyday life. Within the Universal Quantum Foam Hypothesis, this difference is interpreted not as categorical, but as gradual. Every form of field excitation, regardless of whether it is freely propagating or bound, deforms the state of the space field, at least minimally. This continuous deformation constitutes the fundamental source of gravitational effects.

In this framework, gravity does not appear as a phenomenon that only "begins" above a certain threshold, but as a continuous geometric response of the field to every excitation.

Transitions between negligible, weak, and strong gravity do not occur abruptly, but continuously via changes in excitation density, coherence, duration, and the spatial distribution of field-mechanical states. There is no sharp boundary between "non-gravitational" and "gravitational", but rather a continuous spectrum of field-mechanical feedback effects.

The crucial difference between radiation and matter, therefore, lies not in whether a field reaction exists, but in how long, how localized, and how coherent this reaction is. Free radiation generates a mobile, time-limited field deformation that shifts with the propagation of the excitation and relaxes locally. Bound states, on the other hand, fix this deformation spatially and temporally, thereby establishing a permanent stress profile in the field.

From this perspective, matter does not represent a qualitative leap, but rather an extreme case within a continuous transition. It is the state in which field-mechanical deformation is not only generated but also stably stored. Gravitation is consequently not an exclusive attribute of bound mass, but a ubiquitous property of field-mechanical excitation, whose strength, range, and persistence vary smoothly. Thus, gravity does not appear as an abrupt or discontinuous phenomenon, but as a smooth, ever-present reaction of field geometry to dynamics.

5.2 Electrons as a Minimally Stable Field Organization

Against this background, the electron acquires a special conceptual significance.

Within the framework of the Universal Quantum Foam Hypothesis, the electron is not postulated as the only fundamental particle, but rather interpreted as a minimal, empirically particularly stable, and permanently bound field state. This interpretation arises from the observable fact that electrons occur in all known stable atomic and molecular structures and significantly determine their physical, chemical, and macroscopic properties. The UQSH understands this ubiquity not as a random property of a single lepton, but as an indication that the field pattern associated with the electron represents a field-mechanically favored, robust, long-lived, and broadly universally coupling-capable organizational form of the Qu foam.

In this understanding, spin, charge, and interactions are not independent fundamental properties, but rather expressions of different binding, orientation, and entanglement configurations of this minimal field pattern within more complex field structures. Electromagnetic and gravitational effects arise as continuous field-mechanical responses to oriented or permanently bound electron patterns, without replacing or modifying established experimental descriptions.

This interpretation does not contradict established quantum field theory, which describes electrons as fundamental leptons and precisely quantifies their interactions. Instead, UQSH complements this description with a field-mechanical reading in which stability, binding capacity, and universal presence are not postulated but structurally explained. Other particle species, including quarks, hadrons, and gauge bosons, are not reduced or ontologically negated in this framework, but rather classified as higher-order, context-dependent, or composite field states whose dynamics are effectively described by established theories.

In this approach, the electron's special role is therefore not considered exclusive or absolute, but rather understood as an interpretive consequence of its empirically proven stability and universal coupling capability. This view explicitly leaves open the question of whether further

minimal or comparably stable field states might exist, and avoids any statement that would contradict current experimental findings.

5.3 Annihilation as Field Relaxation of Complementary States

Processes such as electron-positron annihilation are described in the Standard Model as the conversion of mass into radiation. The UQSH interprets the same process in field mechanics. Here, the electron and positron are not considered ontologically separate particle species, but rather complementary orientations of the same stable pattern.

When both meet in a suitable configuration, their opposing field orientations allow for a complete relaxation of the bound state. The previously stored field work is thereby transformed into freely propagating radiation modes, typically in the gamma spectrum. Crucially, no "annihilation" takes place, but rather a change of state: from bound, locally supported dynamics to free, propagating dynamics.

Within the framework of UQSH, gamma photons play a special role. They are understood not only as end products of the relaxation of bound states, but also as the primary carriers of those high-frequency, highly focused field excitations from which stable electron patterns can arise under suitable geometric conditions. This does not imply continuous or spontaneous matter creation, but rather describes an extremely rare, geometrically highly selective binding process. In this view, electron formation does not represent an elementary act of creation, but a highly selective binding process of free, high-energy radiation modes.

Gamma radiation thus functions as a field-mechanical boundary mode between the free and bound organization of excitations in the Qu foam: In suitable angular, phase, and density configurations, it can transition into a stable, locally supported electron state; with complementary coupling and relaxation, this bound state in turn dissolves into freely propagating gamma modes. Electrons and gamma radiation thus do not constitute separate ontological categories, but rather two complementary forms of organization of the same field-mechanical dynamics.

5.4 Scale Dependence of Dynamics, Time, and the Speed of Light

(Interpretive Section – UQSH)

The UQSH posits that physical dynamics are not absolutely scale-invariant, but rather remain bound to the specific field-mechanical state space in which stable patterns exist. Quantities such as time, orbital velocity, and the speed of light are therefore not understood as contextless universal constants, but rather as internal limits of the dynamics within a given field regime.

The local invariance of the speed of light, experimentally confirmed in established physics, is fully preserved within the framework of UQSH. For all observers belonging to the same field-mechanical regime, the speed of light is constant, regardless of source, motion, or location. However, UQSH adds an interpretive layer to this statement: The speed of light describes the maximum reaction, orbital, and reorganization rate of stable patterns within a specific field state space. It therefore does not necessarily represent an absolute limit across all possible field regimes, but rather a regime-internal dynamic limit.

A field regime is not primarily understood as a scale of magnitude, but rather as a qualitatively defined state of saturation and organization of the Qu foam. Within each such regime, stable patterns exist—such as electrons, atomic structures, macroscopic bodies, or extremely saturated states—whose internal dynamics are limited by field-mechanical boundary conditions. In this view, time does not arise as a presupposed background dimension, but rather as a measure of the internal orbits, oscillations, and reorganization processes of these stable patterns.

The electron plays a special role in this context. It represents a minimally saturated, yet maximally dynamic, stable field node. Its internal structure allows for high-frequency, continuously sustained orbits that define the finest temporal resolution within the respective field regime. For observers within the same regime, these orbits appear as regular dynamics limited by the speed of light. From the perspective of a different field regime, however, these processes are not measurable as "extremely fast," but rather appear as a stable field state that cannot be further resolved in time.

This relativity of temporal resolution also applies in the reverse direction. Processes occurring in higher saturated field regimes—such as large-scale gravitational stress redistributions or extremely compressed states—appear extremely slow or nearly static from the perspective of lower saturated regimes. This apparent slowness, however, does not indicate a low absolute velocity, but rather reflects the fact that the relevant dynamics are transformed into degrees of freedom that lie outside the temporal resolution of the observing regime.

A direct comparison of velocities or time measures between different field regimes is therefore not physically meaningful. The UQSH does not assume a linear nesting of reference frames, but rather qualitatively different organizational states of the same field. Dynamic quantities are invariant within a regime, but between regimes they are only indirectly accessible via transition and coupling phenomena.

5.5 Electrons and Black Holes as Scale-Dependent Limiting Cases

Within the framework of UQSH, a black hole represents the opposite limiting case to the electron. It corresponds to a maximally saturated field state in which the local field voltage reaches its upper carrying level. Internal cycles of stable patterns can only occur in a highly restricted form in this state; dynamic processes are predominantly converted into carried field voltage.

This does not mean a complete standstill or an abrupt "switch-off" of dynamics. Rather, increasing field saturation is accompanied by a continuous transition in which cycles, oscillations, and reorganization processes are gradually slowed down, rearranged, and finally converted into quasi-static voltage states. Time does not ontologically disappear in this process, but rather loses its physical meaning as a measure of dynamic change, since there are no longer any internal processes that could "tick" in a resolvable form.

For external observers, this state manifests as extreme time dilation, an approach to standstill, and an asymptotic "freezing" of the dynamics. Crucially, this transition is continuous and does not represent a singular break. Extreme time dilation is therefore not an exceptional phenomenon, but rather the limiting case of the same field-mechanical dynamics that proceeds regularly in less saturated regimes.

From this perspective, electrons and black holes do not appear as fundamentally different entities, but rather as scale-dependent realizations of the same field-mechanical stability principle. Both represent locally stable, self-supporting configurations of the Qu foam, in which field tension and dynamics exist in a defined equilibrium. The difference between them lies not in the underlying mechanism, but solely in its degree of saturation and spatiotemporal scale.

The electron represents a minimally saturated, highly dynamic field node in which the field voltage is just sufficient to support a stable structure without suppressing internal processes. Its stability relies on the fact that dynamic processes occur continuously and maintain the field in an activated, temporally highly resolved state. Time is maximally differentiated here, as internal processes occur in fine, reproducible intervals.

In contrast, the black hole marks the maximally saturated limiting case of the same principle. Here, too, a stable field state exists, but the local field voltage is so elevated that internal dynamics are largely converted into supported voltage. Dynamics are not destroyed in this process, but rather bound. The possibility of high-frequency re-emission remains, in principle, as a limiting and unloading phenomenon.

Both states are endpoints of a continuous field-mechanical spectrum in which the ratio of dynamics to tension shifts continuously. There is no sharp dividing line between these regimes, but rather a gradual transition across many scales.

In this sense, the electron and the black hole are not merely analogous, but homologous: They are based on the same structural logic of the field and differ only in the degree of saturation at which dynamics can be sustained. The electron is the smallest possible stable field node in which dynamics dominates; the black hole is the largest possible stable field node in which tension dominates. Both represent stable fixed points of the same field-mechanical organizing rule.

5.6 Scale Relativity of Organization and Observation

(Interpretive Section – UQSH)

The UQSH introduces a scale relativity of organization, according to which physical entities do not exist independently of scale, but always appear as stable organizational forms of a dynamic field medium (Qu foam) within a specific field-mechanical regime. The term "scale" here does not denote an additional space or dimension, but rather a characteristic state space of the field, distinguished by a specific degree of saturation, a limited dynamic resolution, and defined stability windows.

Within this framework, the terms "particle" and "wave" are not ontologically separate categories. Instead, they represent observer-dependent descriptions of the same underlying field state. Whether a field state appears localized or extended depends on which of its internal degrees of freedom are temporally and spatially resolvable within the regime under consideration.

In the context of UQSH, an observer is always part of such a field-mechanical regime. Their measuring instruments, reference times, and spatial resolutions are limited by the dynamics and saturation of this regime. Processes whose internal cycles run faster or slower than the observer's temporal or spatial resolution appear either time-averaged, quasi-static, or as a

continuous field distribution. It follows that the same field-mechanical organization can be described as a "particle," "wave," or "structure" depending on the observation regime, without the underlying physical process itself changing.

UQSH assumes that stable field nodes represent homologous organizational forms across different scales. Homology here means structural similarity of the stability conditions, not identity of the physical parameters or orders of magnitude. Stability arises in every regime through the same fundamental logic: limited field elasticity, nonlinear feedback, geometric closure, and saturation.

From this perspective, extremely saturated field states, such as black holes in observable cosmology, can be understood as limiting cases of stable field organization. They represent locally self-supporting configurations in which dynamic degrees of freedom are largely converted into sustained field tension. The UQSH (Unified Field Structure) allows for the interpretive assumption that such limiting states could assume the role of minimally stable field nodes in a different organizational regime, analogous to the role played by electrons in the known matter regime. This analogy is explicitly to be understood structurally and implies neither an identity of the objects nor a transfer of specific physical properties.

A recurring feature of stable organization is triadic coupling. Field excitations that are coupled exclusively in pairs tend either to propagate or to mutual annihilation. Only triple coupling allows for a closed, rotatable, and self-supporting field structure. This triadic minimum geometry is not restricted to a specific scale, but is interpreted as a general binding condition that can manifest itself differently in different regimes – from elementary bound states to preferred node and junction structures in large-scale field configurations.

In UQSH, time is not understood as a universal background dimension, but rather as a measure of internal dynamics within a field regime (see Section 3.5). The scale relativity of organization therefore does not imply the abolition of time, but rather a regime-dependent limitation of its physical resolution.

In this sense, the scale relativity of UQSH does not describe a new physics, but a new perspective on order. It does not alter the local invariance of the speed of light, nor the formal equations of established theories. Instead, it offers an interpretive framework in which wave-particle duality, time dilation, the stability of elementary matter, and extreme cosmic states can be understood as different manifestations of the same field-mechanical organizing principles.

The scale relativity formulated here remains deliberately open and testable. It makes no claim to ontological finality, but serves the systematic exploration of structural relationships between physical regimes of different scales.

5.7 Scale-Homologique Field States and Interpretive Analogies

(Extended Section – UQSH)

Building on the scale relativity of organization, UQSH allows for a systematic comparison of field-mechanical states across different regimes. This comparison is not to be understood as an equation, but rather as a structurally homologous analogy: What is compared are not objects, but rather stability conditions, coupling logics, and organizational roles within the Qu foam.

5.8 Limiting Regimes of Free Excitation: CMB and Smallest Stable Structures

Within the framework of UQSH, the cosmic microwave background radiation (CMB) represents an extremely frequent, long-lived limiting regime of free field excitation. It is characterized by low excitation density, high coherence, and minimal local field tension. Its stability arises not from binding, but from statistical frequency and field-mechanical relaxation.

These properties allow for a structural analogy to the smallest, weakly bound organizational forms in other regimes. At higher saturated scales, comparable states can appear as frequently occurring, short-lived, or easily reconfigurable structures that prestress, order, or stabilize the field without themselves forming dominant nodes.

The UQSH therefore allows the interpretive assumption that CMB-like states fulfill a stabilizing background function in the respective regime: They modulate field lines, promote couplings, and increase the probability of local organization without themselves being primary bonding nuclei. This role is functionally comparable to weakly bound components in chemical or structural systems.

5.9 Localized Fixed Points: Electrons and Black Holes

In the known matter regime, electrons appear as minimally stable, permanently supported field nodes. Their stability arises from a specific geometric and dynamic balance: sufficient field potential for self-binding while simultaneously preserving internal dynamics. They mark the first stable fixed point of bound organization within this regime.

Black holes represent the opposite limiting case in the cosmological regime. They are maximally saturated fixed points in which additional excitation can no longer be converted into internal cycles but is predominantly stored as supported field potential. Crucially, stability also exists here: not through dynamics, but through saturation.

UQSH allows these two states to be interpreted as endpoints of the same organizational logic. What changes across the scale is not the principle, but the resolution with which dynamics can be supported. Electrons and black holes are thus not opposites, but complementary fixed points of a continuous field-mechanical spectrum.

5.10 Guiding Structures and Coupling Spaces: Filaments as Field Organization

Large-scale filaments and node structures exhibit a pronounced directional order, preferred junctions, and stable connecting axes. Within the framework of UQSH, they are not interpreted as random results of matter distribution, but rather as visible manifestations of field-mechanical guiding structures.

These guiding structures fulfill the same function in their respective regimes as field lines, orbitals, or coupling spaces at lower scales: They direct excitations, favor triadic couplings, and define preferred organizational pathways. In this view, filaments are not objects, but

dynamically supported patterns of order that enable binding without themselves being bound nodes.

5.11 Triadic Coupling as a Cross-Scale Stability Criterion

A central connecting element across all considered regimes is triadic coupling. In UQSH, it is understood as the minimum geometric condition for stable organization. Pairwise couplings remain unstable or transitive; only triple couplings allow for closed, rotatable, and self-supporting patterns.

This structure is found at different levels:

- in elementary matter formation,
- in the organization of electromagnetic states,
- in the preferred geometry of nodes in large-scale field structures.

The recurrence of this geometry is not a numerological finding, but rather an expression of a field-mechanical necessity: stability requires a minimum number of independent degrees of freedom that stabilize each other without collapsing or propagating.

5.12 Defining the Analogy

The analogies described here are not to be understood as literal identities. Galaxies are not photons, and black holes are not electrons in the experimental sense. The UQSH does not claim a direct translatability of physical parameters between scales.

The only things compared are:

- organizational logic,
- stability conditions,
- coupling geometries,
- roles in the overall field-mechanical framework.

The analogy ends where concrete dynamical equations, interaction strengths, or experimental signatures are involved. Within this clear delimitation, it serves as a heuristic tool for the systematic classification of complex structures.

5.13 Integration into the Overall Approach

This scale-homologous approach does not make UQSH more speculative, but rather more structured. It does not replace any theory, but expands the interpretive scope of existing descriptions. Phenomena such as background radiation, filament formation, elementary stability, and extreme cosmic states no longer appear in isolation, but rather as interconnected manifestations of a unified field-mechanical organizing principle.

5.14 Electrical and Magnetic Order Across Scales

(Interpretive Section – UQSH)

Within the framework of the Universal Quantum Foam Hypothesis, electrical and magnetic effects are not understood as separate fundamental forces, but rather as different

manifestations of the same field-mechanical order dynamics. They describe how stable patterns in the Qu foam orient themselves, couple, and mutually influence each other.

5.15 Electrical Order as Local Orientation of Bound Field Nodes

In UQSH, electric charge is not interpreted as an intrinsic substance quality, but rather as an expression of a local orientational asymmetry of stable field nodes. An electrically charged state corresponds to a preferred direction or phase of the field organization, which determines how that node responds to other excitations.

Electrical effects therefore act primarily locally and bindingly. They determine whether field nodes attract, repel, or remain in stable configurations. In atomic and molecular structures, this form of order is dominant because it utilizes finely resolvable degrees of freedom and allows for highly selective bonding.

5.16 Magnetic Order as Large-Scale Coherence of Field Motion

In UQSH, magnetism is understood as a secondary, but ordering consequence of moving or rotating field structures. It does not primarily describe binding, but rather alignment and coherence over larger distances.

Magnetic effects arise where many bound or propagating excitations couple and synchronize their orientations. This results in the formation of preferred axes, field lines, and directional orders that can remain stable even when individual bonds change or break.

At higher scales, magnetism assumes a role similar to that of electrical order at lower scales: It creates structural alignment along which further couplings preferentially develop.

5.17 Scale Transfer of Order Roles

The UQSH allows for a cross-scale assignment of the following functions:

- Electrical order: local stabilization, fine-structured binding, high selectivity
- Magnetic order: large-scale alignment, coupling of many units, guiding structure

On cosmological scales, these roles manifest in the alignment of galactic spin axes, jets, filament directions, and large-scale field patterns. These phenomena are not considered random byproducts, but rather expressions of the same ordering principles that enable binding on smaller scales.

Magnetic fields thus function as structuring guiding geometries along which material and field-mechanical processes preferentially organize themselves. They are the scaled-up equivalent of those ordering forces that enable chemical stability in the atomic regime.

5.18 Interaction with Triadic Geometry

Electrical and magnetic order do not act in isolation, but rather interact with one another. Triadic configurations allow for the simultaneous fulfillment of local binding (electrical) and higher-level alignment (magnetic). Only this combination makes long-lasting, self-supporting structures possible.

Thus, electrical and magnetic effects are not competing forces, but complementary modes of order within the same field, whose relative importance shifts with the scale.

5.19 Entanglement as Non-Local Field Coherence

(Interpretive Section – UQSH)

The UQSH interprets quantum entanglement not as a special phenomenon or additional interaction, but as a fundamental form of organization of coherent field states. In this context, entanglement does not describe an exchange of information across distances, but rather the fact that multiple spatially separated measurement events access the same physical field state.

5.20 Separate Locations, Non-Separated States

A central misunderstanding of entanglement is to automatically equate spatial separation with state separation. In UQSH, a strict distinction is made between:

- the spatial localization of measurement events and
- the spatial extent or coherence of a field state.

Two entangled particles can be located at spatially widely separated locations without the underlying field state itself being spatially fragmented. The state is not localized "between" the locations, but rather encompasses them as a coherent unit. Entanglement therefore does not mean action at a distance, but rather shared state membership.

5.21 Field-Mechanical Description of Entanglement

Within the framework of UQSH, entanglement arises when field excitations do not relax as independent local states, but rather retain a common coherent organization. As long as this coherence is not resolved field-mechanically, there is no physical meaning in assigning the state to individual locations.

In this understanding, measurements do not access a "hidden state," but rather locally enforce a state fixation within a previously global field mode. The correlations between measurement events do not result from a subsequent exchange of information, but from the fact that both measurements partially fix the same pre-existing field state.

This explains why entangled systems exhibit correlations that cannot be explained classically without violating the local causal structure or the invariance of the speed of light. No signal transfer takes place; rather, an already shared state is locally specified.

5.22 Magnitude and Spatial Range

From the perspective of UQSH, the entangled state itself does not possess a characteristic length corresponding to the distance between the measurement points. The spatial separation of the measurement points can range from microscopic to macroscopic scales without altering the physical status of the state.

The state does not occupy any additional space "between" the locations. It is not an object, not a wave in the classical sense, and not a medium with a localizable extent. Rather, it is a

coherent organizational form of the field whose spatial range is determined not by geometric extent but by coherence maintenance.

5.23 Measurement as Local State Fixation

In UQSH, the measurement process is not interpreted as a mysterious collapse, but rather as a local field-mechanical fixation of a previously unlocalized state. The measurement does not instantaneously change a distant location, but rather reduces the degrees of freedom of the shared field state.

The fact that a second measurement yields consistent results is not a consequence of an interaction between the measurement locations, but rather an expression of the fact that both measurements are part of the same state space. The temporal order of the measurements plays no fundamental role, since no causal influence in the classical sense takes place.

5.24 Differentiation from Misinterpretations

The field-mechanical interpretation of entanglement within UQSH explicitly implies:

- no faster-than-light communication,
- no violation of local causality,
- no additional forces or fields,
- no physically transported information between measurement points.

Entanglement is neither a signaling mechanism nor an action at a distance, but a property of coherent field organization. The mathematical predictions of quantum mechanics remain completely intact; only their ontological interpretation is altered.

In this context, quantum entanglement provides consistent empirical evidence for the relationality of physical states in space. Entangled systems exhibit nonlocal correlations without signal transmission or violation of local causality.

Within the framework of UQSH, this behavior can be interpreted as an expression of a shared field state, where spatial separation is not equivalent to field-mechanical independence. Entanglement thus does not point to a cancellation of space, but rather to its field-based, relational structure, in which states remain connected via the medium, even if they appear geometrically separate.

5.25 Transition to Field Organization and Matter Formation

In this context, entanglement represents a preliminary stage of field-mechanical organization. As long as a state is coherent but not geometrically closed, it remains entangled and not locally bound. Only when additional conditions are met—such as geometric closure, triadic coupling, or sustained field saturation—can an entangled field excitation become a bound, locally stable state.

Matter thus appears not as the opposite of entanglement, but as its field-mechanically stabilized form. Entanglement is not an exotic exception, but the normal case of coherent field excitation; material binding is the highly selective special case.

5.26 Classification

The interpretation of entanglement presented here is not intended as a replacement for established quantum mechanics, but rather as a conceptual framework that embeds its formal results in a consistent field-mechanical ontology. It explains the non-locality of quantum mechanical correlations without resorting to hidden variables, action at a distance, or ontologically problematic assumptions, and fits seamlessly into the overall approach of the Universal Quantum Foam Hypothesis.

5.27 Transition from Entanglement to Matter Formation

(Interpretive Section – UQSH)

Within the framework of the UQSH, matter formation is not understood as a categorically new physical process, but rather as a specialization and stabilization of field-mechanical coherence. The transition from entanglement to matter does not mark a break between quantum mechanical and classical descriptions, but rather a continuous regime change within the same dynamic field medium.

In this context, entanglement does not describe an exotic special state, but a general property of coherent field excitations: degrees of freedom are not independently localized, but are relationally organized across the field. Such entangled states can be large-scale, dynamic, and time-varying without forming a fixed spatial constraint. Entanglement thus represents a necessary, but not sufficient, condition for matter formation.

Matter only arises where this field-mechanical coherence can no longer be freely reorganized or relaxed, but instead becomes locally fixed. This transition does not occur through an ontological collapse, but rather through a field-mechanical stabilization of specific organizational modes. Free field excitations gradually transition from globally distributed coherence to locally bound self-coupling. In this view, matter formation is the development of a self-supporting field node in which dynamic processes continue to occur, but no longer propagate outwards.

A central element of this transition is the geometric selectivity of the stabilization. The UQSH assumes that permanent binding is only possible if multiple field excitations interact within a nonlinear, closed coupling geometry. In particular, triadic constellations allow for a rotatable, self-stabilizing organization, since pairwise couplings tend towards transmission or annihilation. This minimum triadic geometry is not a numerical assumption, but follows from the minimum condition for nonlinear feedback without external fixation.

The resulting bound states have no sharp spatial boundaries. This lack of clarity is not a deficiency, but a constitutive property of stable matter. The finite extent, internal dynamics, and lack of precise localizability allow the field to bear tension without collapsing into divergences. Matter is therefore not stable despite this lack of clarity, but precisely because of it.

The transition from entanglement to matter occurs continuously:

- freely propagating, globally correlated field excitation
- geometric focusing and condensation
- self-coupling of certain degrees of freedom
- formation of a locally bound, self-supporting field node

No violation of local invariances occurs. In particular, the experimentally confirmed statements of quantum mechanics and relativity remain fully intact. The UQSH provides an interpretive framework in which matter is understood as a permanently organized limiting case of field-mechanical coherence.

Matter formation thus appears not as a singular event, but as a highly selective, dynamic stabilization process within a continuous field system. Entanglement and matter are not opposing concepts, but rather different degrees of organization of the same physical medium.

5.28 Scale-Homologique Assignment of Organizational Roles

(Structuring Overview – UQSH)

For better understanding, the previously described states can be functionally contrasted. This assignment is interpretive and describes roles, not identities.

Free Field Excitation / Background States

• Local State:

Weak, long-wavelength radiation with limited spatial coherence

→ acts locally modulating without defining large-scale field geometry

• Large-Scale State (CMB):

Globally coherent, extremely long-wavelength circulating state of free field excitation

→ acts as field-mechanical prestressing, background stabilization, and large-scale modulation of the Qu foam

Function within the UQSH framework:

The CMB does not represent an active dynamic process, but rather a long-lived background state of minimum excitation density that supports the large-scale field geometry, limits structural relaxation, and field-mechanically modulates other radiation and matter processes.

Minimally stable field node

- Low scale: Electron
- Higher scale: Extremely compact, long-lived field node
- Function: First stable binding point

Conducting and coupling structures

- Low scale: Orbitals, field lines
- Higher scale: Filaments, jets, large-scale field axes
- Function: Alignment, coupling, preferred organization

Maximally saturated field node

- Low scale: Theoretical limiting case of strong binding
- Cosmological scale: Black hole
- Function: Stress absorption, limiting stability

This overview is not intended to reduce complex phenomena, but rather to provide structural orientation within the overall model.

5.29 Clear Delimitation of the Analogy

(Criticism-Proof Classification – UQSH)

The scale-homologous analogies used in UQSH are deliberately limited. They do not replace established physics and do not claim a direct translatability of quantities, forces, or equations between scales.

In particular:

- Galaxies are not photons.
- Black holes are not electrons in the experimental sense.
- Filaments are not field lines on the electrodynamic scale.

Only the following are compared:

- Stability conditions,
- Organizational logics,
- Coupling geometries,
- Roles in the overall field-mechanical structure.

The analogy ends where concrete measurable quantities, dynamic equations, or experimental predictions are involved. It serves as a heuristic tool to reveal structural relationships, not as an assertion of physical identity.

Within this clear boundary, the analogy strengthens the conceptual coherence of the UQSH without making it empirically vulnerable.

5.30 Black Holes as Saturated Limit States of the Field

Within the framework of the UQSH, black holes are not interpreted as singular breaks in physics, but rather as maximally saturated, locally stable limit states of the same field-mechanical medium from which radiation, matter, and gravity also emerge. They do not mark a breakdown of the field description, but an extreme case of continuous field organization.

In a black hole, the local field tension of the Qu foam is so greatly increased that additional excitation can no longer be primarily converted into internal orbits, oscillations, or reorganizing dynamics. However, this does not mean a complete cessation of all processes. Rather, the relationship between dynamics and tension shifts: Dynamic degrees of freedom are increasingly suppressed and converted into permanently sustained field tension. Internal processes continue, but with extremely slowed dynamics that are practically no longer temporally resolvable for external observers.

Time does not ontologically disappear, but rather loses its physical function as a measure of internal change. Within the UQSH framework, time is always bound to dynamic cycles. Where these cycles have almost completely transitioned into field tension, there are no longer any internal processes by which a temporal sequence could be measured. For external observers, this manifests as extreme time dilation, asymptotic slowing down, and an apparent "freezing" of the dynamics. This transition is continuous, not abrupt.

A black hole thus represents the opposite limiting case to the electron. While the electron is a minimally saturated, highly dynamic stable field node, the black hole represents a maximally saturated, almost dynamic-free stable field state. However, these are not different principles, but rather scale-dependent realizations of the same field-mechanical stability rule. The difference lies solely in the degree of saturation and the spatiotemporal scaling, not in the underlying physics.

Crucially, black holes in the UQSH framework do not represent energy sinks. Incoming matter and radiation are not annihilated, but rather transformed into highly saturated field stress over long periods. This stress does not remain completely static. Even in maximally saturated states, a residual dynamic exists that allows bound patterns to be reorganized under extreme pressure.

This reorganization is not isotropic. Due to rotation, accretion geometry, and large-scale field structure, field-mechanically favored release channels form, along which the effective counter-elasticity of the field is minimal. These channels manifest astrophysically as jets. Jets are not ejections in the classical sense, but rather directions of minimal field-mechanical impedance along which accumulated field stress cannot be carried further and therefore inevitably transitions into free dynamics.

Within the context of ultra-high quantum shock (UQSH), gamma-ray bursts (GRBs) do not occur as initial events or primary energy sources, but rather as late, highly concentrated discharge events within long-term field-mechanical evolution processes. They do not mark a beginning, but rather the endpoint of a local field concentration built up over long periods.

Large-scale accumulations of matter, particularly in strongly curved or prestressed field regions, lead to a gradual accumulation of field stress in the quantum foam. This stress is not continuously reduced, but can be maintained in a bound or focused form for very long periods. The dynamics of these states are greatly slowed, but not completely eliminated.

When reorganized field excitations or bound states of matter finally reach a field-mechanical relief zone—for example, along preferred directions of minimal field stress or geometrically focused transition regions—the previously stored field work can no longer be propagated. At this moment, a rapid conversion of the bound or highly concentrated field stress into free, high-frequency radiation modes occurs.

The gamma radiation released in this process does not represent energy generation, but rather the return of previously stored field work to freely propagating excitation. The extreme short duration, high energy density, and strong collimation of GRBs result from the collective, nearly simultaneous discharge of a long-accumulated stress state, and not from a singular, explosive process of matter.

In this view, GRBs appear as point signatures of a continuous, field-mechanically stored evolutionary process. Their time lag relative to the underlying matter accumulation is not

accidental, but rather an expression of the slow internal dynamics of highly saturated field states and the abrupt nature of their release once a critical transition is reached.

Hawking radiation can be understood in this context as the weakest limiting case of the same field-mechanical re-emission. Here, too, it is not an internal dynamic process of the black hole, but an unavoidable, minimal redistribution of field stress at the boundary of maximum saturation. The thermal, isotropic, and extremely weak nature of this radiation results directly from the near-complete suppression of internal dynamics.

If one considers a black hole not as an exotic exception, but as a field-mechanical state with extremely slowed internal dynamics, the scale-dependent nature of physical processes becomes immediately apparent. Crucially, electrons and black holes are not described by different principles, but by the same mechanism, albeit with vastly different degrees of field saturation.

The electron represents a minimally saturated, highly dynamic field state. Its stability relies on the fact that internal orbits, oscillations, and reorganization processes can be sustained indefinitely without the field tipping into a saturated stress state. These internal processes proceed with maximum possible dynamics within the respective field regime. Time is highly resolved here, as state changes occur in minute, reproducible intervals. For observers of this regime, this dynamic appears normal, limited by the speed of light, and continuous.

The black hole represents the same mechanism at the opposite end of the scale. Here, the field state is near its maximum saturation, so that additional excitation can hardly be converted into internal dynamics. Orbits, oscillations, and reorganization processes continue, but at an extremely slow rate and increasingly transformed into sustained field tension. To external observers, these processes appear practically timeless, as their timescales far exceed the resolution of the observing regime.

The crucial point is:

The differences between electrons and black holes do not arise from different physical laws, but from the relative speed at which field states can realize their internal dynamics. What appears as fast, finely resolved dynamics in electrons manifests as extremely slow, stress-dominated evolution in black holes. Both are stable endpoints of the same field-mechanical organizing principle. The scale does not determine what happens, but rather how fast and how resolvable it is.

This explains why black holes appear "slow" to us, while electrons appear "fast." These attributions are not absolute properties, but observer- and scale-dependent projections of the same underlying field behavior. An observer belonging to a more saturated regime would perceive the dynamics of an electron not as fast, but as stable and not further differentiated in time—analogous to how we perceive the dynamics of a black hole.

The scales are thus not explained by additional dimensions or new constants, but by the saturation level of the field, which determines how much dynamics can be carried per unit field volume. Electrons and black holes are not opposites in the ontological sense, but rather two reference points where it becomes apparent how dynamics continuously transform into field tension.

This is precisely why cross-scale phenomena can be conceived consistently in UQSH without abandoning local invariances (such as the speed of light):

Each scale possesses its own internal dynamics, its own temporal resolution, and its own stable patterns—but all are based on the same field.

In summary, black holes in the UQSH appear as stable saturation nodes of the Qu foam, in which dynamics are largely converted into field tension, but never completely disappear. Discharge, re-emission, and restructuring remain possible in principle, but occur highly selectively, anisotropically, and over long field-mechanical timescales. Black holes thus do not represent an endpoint of physics, but rather a consistent limiting case of a continuously dynamic, energy-conserving field system.

5.31 Gravitation as a Continuous Field Response

In this understanding, gravity arises as the immediate geometric response of the Qu foam to any form of field deformation. Both free radiation excitations and bound matter patterns locally alter the field state, thereby generating a gravitational effect, which is generally extremely weak. The crucial difference lies not in the existence of gravity, but in its temporal and spatial integration.

Free radiation causes dynamic field deformations that propagate along the field and relax rapidly. Bound patterns, on the other hand, prevent the complete relaxation of the field and thus generate permanently load-bearing stress states. Gravitation is therefore not a threshold or binding phenomenon, but a constantly present property of field dynamics, whose macroscopic effectiveness results from the persistence and superposition of field deformations.

5.32 Unified Field-Mechanical View of Matter, Radiation, and Gravitation

In summary, the UQSH describes matter, radiation, and gravity as different forms of organization and manifestation of the same dynamic field medium. Electrons function as minimally stable, locally bound field states, free radiation as propagating, non-permanently bound excitation, and gravitational effects as a continuous geometric response of the field to both forms of organization. Scales, timescales, and dynamic limits do not arise absolutely, but rather from the respective saturation and organization levels of the field.

This results in a coherent and consistently interpretable conceptual framework that neither replaces nor modifies known phenomena of quantum physics, relativity, and cosmology, but rather reclassifies them within a common field-mechanical language, structurally connects them, and ontologically unifies them.

5.33 Electromagnetism as Flexible Coupling of Bound Patterns

Electromagnetic interactions are interpreted as structured field excitations of the Qu foam that preferentially couple to bound field nodes. Electric charge corresponds to an orientation- or phase-dependent asymmetry of stable pattern nodes.

This classification explains the special role of electromagnetism as the most flexible and longest-range interaction at the atomic and molecular scales. Since the electron is understood in the UQSH framework as a minimally stable, universally coupling pattern node,

electromagnetism constitutes the dominant binding and organizing regime of chemistry and macroscopic material structures.

5.34 Strong Interaction as a Stiffness and Saturation Regime

From a field mechanics perspective, the strong interaction is interpreted as a regime of highly bound, locally extremely dense patterns. Protons and neutrons do not appear as elementary objects, but rather as complex stress nodes with internal modes. These internal degrees of freedom are described in the Standard Model by quarks and gluons.

In the UQSH framework, the strong interaction corresponds to a bound state near the local stress or elastic limit of the field. The resulting patterns are highly stable, spatially small, and extremely stiff with respect to external perturbations. Their binding is not flexibly scalable and therefore has no macroscopic range. In this sense, the strong interaction does not function as a universal organizing force, but rather as an internal stabilization mechanism for extremely dense field states.

5.35 Weak Interaction as a Rearrangement and Relaxation Channel

The weak interaction is interpreted as a field-mechanical rearrangement and relaxation channel through which unstable or marginally stable pattern configurations can transition into other allowed states. It does not represent a binding mechanism, but rather describes rare, yet allowed, reorganization processes within the pattern space.

Decays and identity changes of particles are understood in this context as field-mechanically consistent rearrangements of bound patterns, in which conserved quantities are maintained while the internal pattern structure changes.

5.36 Back-Binding to Electron Primacy and 3×3 Selectivity

The UQSH assumes that only a very limited class of stable bound field patterns exists. The formation of permanent states of matter is subject to a geometric selectivity described by a minimal complete constellation of three functional roles, three geometric orientations, and three energy levels (3×3 structure).

Within this framework, the electron represents the simplest completely stable pattern node that satisfies these selectivity conditions. Higher-order matter structures do not arise from new fundamental patterns, but rather from the nesting, coupling, and internal modulation of this basic pattern. The different interactions merely reflect the respective binding, stiffness, or rearrangement characteristics of the field states involved.

5.37 Observable Distinguishing Features

This classification leads to fundamentally verifiable distinctions compared to substance-based interpretations:

First, gravitational effects are not necessarily linked to separate matter objects, but can occur as a continuous stress architecture of the field.

Second, stable states of matter are not solely determined energetically, but are also geometrically selective.

Third, field-mechanical saturation explains why bonds at different scales exhibit natural limitations and why certain structures do not condense further despite high energy density.

6. Quantum Physics as Field State Theory (UQSH)

6.1 Locality and Non-Locality in the Qu Foam

Quantum physics describes the behavior of elementary systems, interactions, and measurement processes with high precision. Its mathematical structures are extraordinarily successful, yet the physical interpretation of many concepts—such as wave functions, superposition, entanglement, or collapse—remains open to interpretation. The UQSH is not intended as a replacement for quantum mechanical formalisms, but rather as a field-mechanical classification of their significance.

Within the framework of UQSH, quantum physics is interpreted as a description of local and non-local states of a continuous, dynamic field medium. Quantum objects are not isolated point entities, but rather stabilized patterns, excitations, or coupling states within this field. The quantum mechanical description does not capture "particles per se," but rather the possible states and transitions of field-borne patterns.

6.2 Wave Function as Real Field Information

In UQSH, the wave function is not understood as a mere computational tool or subjective piece of knowledge, but rather as an effective description of the spatially distributed excitation and coupling structure of a field state. Its extent reflects the real non-locality of field-mechanical states. Superposition does not correspond to a paradoxical simultaneity of contradictory properties, but rather to the superposition of several compatible field configurations within the same medium.

The so-called collapse of the wave function is interpreted in this context as a local coupling and stabilization process. A measurement fixes a previously extended field state to a concrete, measurable configuration. No information jump is postulated, but rather a transition from distributed to locally bound field organization.

6.3 Entanglement as Shared Field Coherence

In UQSH, quantum entanglement does not appear as a mysterious action at a distance, but rather as an expression of a shared, coherent field structure. Entangled systems are not separate objects with subsequent correlation, but rather partial states of a connected field state that cannot be completely decomposed into local components.

Nonlocal correlations arise because measurements at different locations sample the same global field state. No information is transmitted faster than light; rather, a pre-existing field-mechanical coherence is made visible. The limitations of signal transmission remain completely intact, while the quantum mechanical correlations acquire a natural geometric explanation.

6.4 Energy, Momentum, and Quantization

In UQSH, the quantization of energy, momentum, and other observables is understood as a consequence of stable field modes. Discrete energy levels correspond to preferred, stable organizational forms of the field under given boundary conditions. Quanta are not "smallest substances," but rather minimal, reproducible state transitions of the field.

This perspective allows us to understand quantization not as fundamentally discontinuous, but as an emergent result of a continuous medium with limited elasticity, stability conditions, and geometric selectivity. Quantum physics thus describes not the breaking of continuity, but its structural self-organization.

6.5 Classification and Scope

The UQSH does not alter any experimentally confirmed predictions of quantum mechanics. All established results—interference, tunneling, spin, entanglement, and the uncertainty principle—remain valid. The only new aspect is the ontological framework: quantum phenomena are no longer considered fundamentally enigmatic, but rather natural expressions of a field-mechanical system.

Thus, quantum physics transforms from a theory of "strange particles" to a theory of field-borne patterns and coupling states. Together with gravity, radiation, and matter, it represents different regimes of the same physical medium. The UQSH therefore understands quantum physics not as a special case, but as a local description of the general field dynamics of the universe.

UQSH Relationship to Established Physics

The UQSH is formulated as a conceptual framework in such a way that it does not contradict established experimental findings in modern physics. It does not replace established theories, but rather offers an alternative field-mechanical interpretation of the physical meaning of observable effects.

In particular, the UQSH is compatible with:

- the local invariance of the speed of light,
- general relativity as a geometric description of gravitational effects,
- quantum field theory with respect to its empirically confirmed predictions of quantized excitations,
- established electromagnetic models.

The deviations primarily occur at the interpretive level. Space is not understood as a passive geometric stage, but as a physical field. Time is not interpreted as an independent fundamental dimension, but as a field-mechanical property of dynamic states. Dark matter and dark energy are not introduced as additional substances, but interpreted as properties of large-scale field states.

The observable effects—including rotation curves, gravitational lensing, time dilation, and quantum mechanical correlations—remain completely intact. Thus, the UQSH does not shift the empirical basis of physics, but rather the ontological interpretive framework within which these effects are interpreted.

Limitations and Open Questions

The UQSH represents a conceptual framework and does not claim to be a complete dynamical theory. Several key aspects are deliberately left open, as the UQSH focuses on a field-mechanical reinterpretation of established phenomena, not on an immediate, complete formalization.

The most important open questions include:

- the development of a quantitative field-mechanical description of the Qu foam,
- the derivation of verifiable predictions regarding the geometric selectivity of matter formation,
- a precise mathematical formulation of the maximum elastic limit of the field.

Furthermore, it is necessary to clarify under which physical conditions experimental or astrophysical evidence for field-mechanical prestresses, relaxation processes, or selective matter formation can be obtained.

In this sense, the UQSH sees itself as a structuring framework that consistently connects known phenomena and specifically extends existing models where they exhibit interpretive gaps. Its strength lies not in providing definitive answers, but in the systematic organization of open questions and the establishment of a coherent field-mechanical reference framework.

Conclusion: What changes – and what doesn't?

The field-mechanical interpretation presented here does not aim to replace or relativize existing observations. Rotation curves, gravitational lensing, substructure in halos, the cosmic web, and the statistical relations between baryonic matter and gravitational effects remain unchanged. What changes is the ontological framework within which these findings are interpreted.

In the standard model, many of the discussed stress fields – radial acceleration relation, core-cusp, too-big-to-fail, missing satellites, or the separation of gas and lens centers in the bullet cluster – are interpreted as evidence of an additional, as yet unproven, substance. In UQSH, these same phenomena are understood as properties of a dynamic space field that can generate, support, and limit stress states. Additional gravitational effects arise not from "more matter," but from the organization, stability, and saturation of field-mechanical patterns.

A key difference lies in the fact that gravitational influence and material manifestation are no longer equated. Field tension can exist without necessarily manifesting as compact baryonic objects. Conversely, visible matter is particularly stable where it is embedded in coherent field-mechanical structures. Many of the classical dark matter problems lose their anomaly character within this framework and appear as different facets of the same fundamental structural principle.

Importantly, this approach does not claim to offer a new particle theory, replace established equations, or provide a complete dynamical formulation. Nor does it claim that dark matter has been "disproven." Instead, it demonstrates that observable phenomena are ontologically ambiguous and that a field-mechanical framework can interpret the same data with fewer categorical assumptions.

The true test of such an approach lies not in its rhetorical elegance, but in its consequences. The UQSH implies that gravitational substructure is predominantly baryonic and geometrically shaped, that halos represent continuous stress profiles without hard object

boundaries, and that the visibility of satellite galaxies strongly depends on the surrounding environment, dynamics, and field geometry. These statements are, in principle, observably falsifiable and differ in their predictions from purely mass-based models.

Whether the framework outlined here is viable ultimately depends not on individual "successes," but on whether, in the long term, it helps to organize known phenomena more consistently, formulate new questions more precisely, and avoid unnecessary ontological assumptions. In this sense, this contribution is not intended as a conclusion, but as an invitation to critical examination: less in the mode of "What's still missing?", but rather in the mode of "What do we actually assume when we say dark matter?"

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The following works serve to contextualize established physical descriptions, upon whose empirical and theoretical foundation the interpretive framework presented here is based. They do not constitute a direct basis for the derivation of the UQSH, but rather mark the conventional context within which the discussed phenomena are usually treated.

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